

Almanac 47 for Free42

SwissMicros DM42 / DM42n with the stock (Free42) firmware — user manual

The full Almanac 47 of the C47 — the Sun, the Moon, the planets and the 58 navigation stars, 9 views — converted to Free42 3.3 with the DM42 graphics extension (the whole 400 × 240 screen). The menu, the screens and the values are the same as on the C47: in the Free42 core (SwissMicros source, release 3.3.10) each screen was compared pixel by pixel with the C47 version, and the text page line by line.

It supports, and does not replace, the Nautical Almanac. Always cross-check the values.

Copyright © 2026 Victor Valdes. Free software under the GNU General Public License v3.0 or later. It comes with NO WARRANTY. Written with the help of AI (Claude, by Anthropic); the results were checked against the C47 version, which was checked against JPL Horizons, USNO data and the Nautical Almanac.

1. What you need

- A **DM42n** (recommended) or DM42 with the SwissMicros stock firmware (DM42 3.26 = Free42 3.3.10 or later). The DM42n has ample memory. For little memory there is NAVLITTLE: the ALMANAC screen with the Sun and the 58 stars only (section 10).
- Free42 on a PC or phone has only the 131 × 16 HP-42S screen: NAV runs there, but you see only the top left corner of each screen. Use the calculator for the views.

2. Files (build/free42)

File	What it is
NAVFULL.raw	the program NAV and its routines: all 9 views; the screen appears complete, at once, as on the C47
NAVINIT_FULL.raw	INIT for NAVFULL: the series matrices, valid 2000–2050
NAVINIT_FAST.raw	INIT for NAVFULL: fitted series for 2026–2030 (smaller, faster)
NAVLITTLE.raw	NAV for little memory: the ALMANAC screen, Sun and stars only (section 10)
NAVINIT_LITTLE.raw	INIT for NAVLITTLE: Sun, nutation and stars, valid 2000–2050
TBL_1.raw, TBL_5.raw	optional almanac tables (JPL) for 1 or 5 years from 1 Oct 2026 (section 11)
*.txt	the same programs as text (Free42 on a PC: Paste in PRGM mode)
dev/NAVFULL_DRAW.raw	NAVFULL with the Free42 screen update: each screen builds up as it is drawn

3. Loading and first start

- Connect the calculator by USB (it appears as a disk) and copy the .raw files into the PROGRAMS folder.
- The file name is the version: on the calculator the programs are always NAV and INIT.
- On the calculator: **SETUP → Load Program**, load NAVINIT_FULL.raw (or NAVINIT_FAST.raw).
- **XEQ "INIT"**: it builds the matrices and shows MATRICES READY. Then delete INIT (GTO "INIT", CLP): the matrices stay.
- Load NAVFULL.raw and **XEQ "NAV"**.

4. Starting NAV

NAV asks four numbers with INPUT (R/S keeps the value shown); the formats are in the message line first:

Input	Meaning	Example
DATE	UT date, YYYY.MMDD	2026.0926
UTC	UT, HH.MMSS	14.57
LAT	latitude DD.MMm, south negative	25.20 = 25° 20' N
LON	longitude DDD.MMm, west negative	55.12 = 55° 12' E

- **Time is UT (UT1)**: for a sight timed in UTC add DUT1 (under 0.9 s) — as in the almanac.
- **Signs**: latitude N +, S –; longitude E +, W –.

Then NAV switches to the 400 × 240 graphics mode (GrMod 3), computes the sky (the SINKING....ABOUT box shows meanwhile) and draws the menu. 0 on the menu ends NAV and sets the normal screen again (GrMod 0).

ALMANAC 47		VALID 2026-2030	
26-09-2026 14:57 UT DR N 25 20.0 E		55 12.0	
1	ALMANAC	6	ANIM
2	CHART	7	ALLSKY
3	TEXT	8	INFO
4	SKY	0	END
5	SPLIT		
KEY A NUMBER + MENU UP DOWN 1 HOUR			
DOES NOT REPLACE THE NAUTICAL ALMANAC			

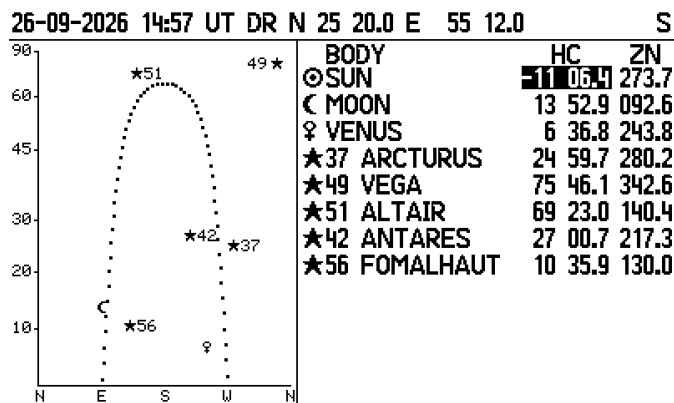
5. Keys

Key	On the menu	On a view
1 – 8	the item is highlighted and its view is drawn	—
9	PRLCD: the screen to the printer (a screenshot)	PRLCD
+	—	back to the menu
▲ / ▼	one hour later / earlier	the same view one hour later / earlier
0	end of NAV (screen and stack cleared)	—
R/S, EXIT	stop the program	stop the program

6. The views

26 Sep 2026 14:57 UT, 25° 20' N 055° 12' E, as drawn by Free42.

26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0					S
	BODY	GHA	DEC	HC	ZN
	ARIES	229 40.8			
☉	SUN	46 25.7	S 1 24.5	11 06.4	273.7
☾	MOON	228 27.2	N 3 37.2	13 52.9	092.6
♀	VENUS	16 40.1	S 20 19.3	6 36.8	243.8
★37	ARCTURUS	15 27.7	N 19 02.7	24 59.7	280.2
★49	VEGA	310 13.1	N 38 48.8	75 46.1	342.6
★51	ALTAIR	291 39.3	N 8 56.5	69 23.0	140.4
★42	ANTARES	341 55.1	S 26 29.5	27 00.7	217.3
★56	FOMALHAUT	244 53.4	S 29 28.7	10 35.9	130.0
NAUT TWI 01:20 15:01			MOON 100% ● FULL		
RISE/SET 02:09 14:12			AGE 15.5 DAYS		
MER PASS 08:11 SD 16.0			HP 57.7 SD 15.7		



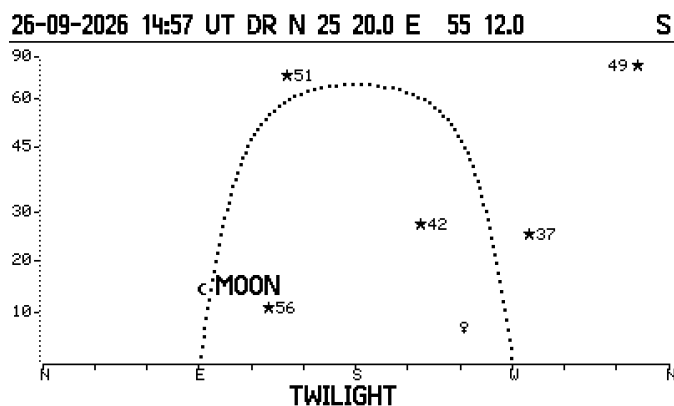
1 ALMANAC: GHA, Dec, Hc, Zn of 10 bodies; twilight, rise/set, mer. pass, Moon phase, HP, SD. A negative Hc is white on black.

2 CHART: the sky (Hc up, Zn across, the celestial equator dotted) and the Hc/Zn table.

26-09-2026 14:57 UT DR N 25°20.0' E 55°12.0'	GHA 310°13.1' DEC N 38°48.8'
ARIES 229°40.8' S	51 ALTAIR HC 69°23.0' ZN 140.4°
★ SUN HC -11°06.4' ZN 273.7°	GHA 291°39.3' DEC N 8°56.5'
GHA 46°25.7' DEC S 1°24.5'	42 ANTARES HC 27°00.7' ZN 217.3°
MOON HC 13°52.9' ZN 092.6°	GHA 341°55.1' DEC S 26°29.5'
GHA 228°27.2' DEC N 3°37.2'	56 FOMALHAUT HC 10°35.9' ZN 130.0°
VENUS HC 6°36.8' ZN 243.8°	GHA 244°53.4' DEC S 29°28.7'
GHA 16°40.1' DEC S 20°19.3'	53 DENEBC HC 61°13.8' ZN 039.3°
SATURN HC 3°32.1' ZN 089.2°	GHA 279°05.4' DEC N 45°22.8'
GHA 217°39.7' DEC N 2°13.1'	NAUT TWI 01:20 15:01 RISE/SET 02:09 14:12
37 ARCTURUS HC 24°59.7' ZN 280.2°	MER PASS 08:11 SUN SD 16.0
GHA 15°27.7' DEC N 19°02.7'	MOON 100% FULL AGE 15.5 DAYS
49 VEGA HC 75°46.1' ZN 342.6°	MOON HP 57.7 MOON SD 15.7

+ MENU

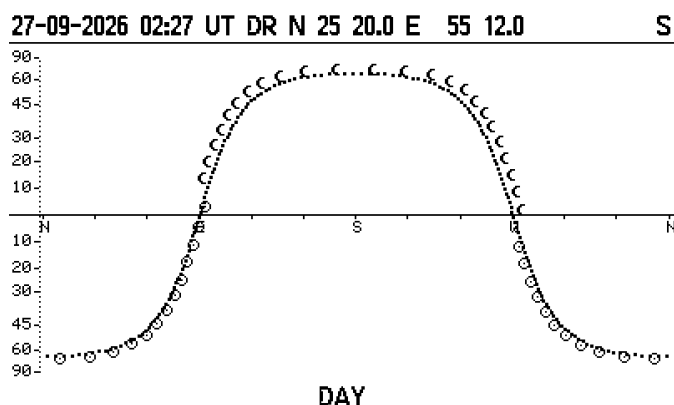
3 TEXT: the almanac page as text, drawn with the small font; the lines are also in R50 ... (Free42 has no register browser). + back to the menu.



4 SKY: full-screen chart; every second the name of the next body is shown next to it; DAY / TWILIGHT / NIGHT at the bottom.

26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0 S

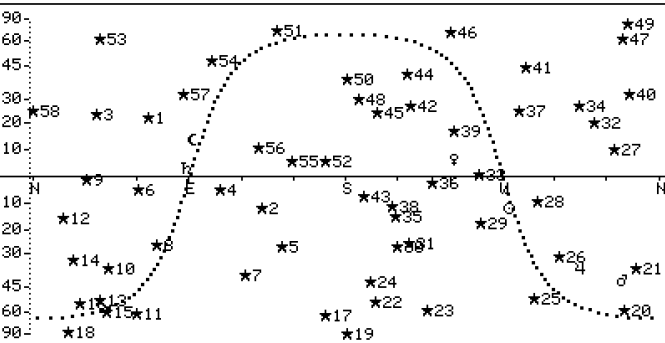
	BODY	GHA	DEC	HC	ZN
☉	SUN	46 25.7	S 1 24.5	11 06.4	273.7
☾	MOON	228 27.2	N 3 37.2	13 52.9	092.6
♀	VENUS	16 40.1	S 20 19.3	6 36.8	243.8
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★56	FOMALHAUT	244 53.4	S 29 28.7	10 35.9	130.0



5 SPLIT: chart on top, the bodies below down to the bottom.

6 ANIM: the Sun and the Moon over 12 hours (24 frames, 1 s each).

26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0 S



TWILIGHT

7 ALLSKY: the whole sky, over and under the horizon; DAY / TWILIGHT / NIGHT.

ALMANAC 47 - INFO

GITHUB.COM/VALDESVJ/ALMANAC47
COPYRIGHT 2026 VICTOR VALDES
FREE SOFTWARE: GNU GPL V3 OR LATER

NO WARRANTY - IT MAY HAVE BUGS AND
ALWAYS CROSS-CHECK THE VALUES
WITH THE NAUTICAL ALMANAC

THE ANTS ARE WAITING FOR A FLAG
+ MENU

DOES NOT REPLACE THE NAUTICAL ALMANAC

8 INFO: repository, licence, no warranty, cross-check (the warning is on the menu and INFO only).

ALMANAC 47VALID 2026-2030
26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0

1 ALMANAC

2 CHART

3 TEXT

4 SKY

5 SPLIT

6 ANIM

7 ALLSKY

SINKING....ABOUT

0 END

KEY A NUMBER + MENU UP DOWN 1 HOUR

The SINKING....ABOUT box while the calculator works, with the ants (flag 97, below).

7. How the screen appears

Free42 sends every drawing step to the LCD at once, so a program builds each screen up in front of you (the stars of SKY appear one by one), and the SINKING box is cleared as soon as the next view starts to draw. The C47 shows its screen only at a PAUSE, a key press or the end, so there a view appears complete, at once, and the box stays until it is ready.

NAVFULL and **NAVLITTLE** do it the C47 way with the DM42 variable RefLCD: 0 STO "RefLCD" (no LCD update while NAV computes and draws), -1 STO "RefLCD" (one update) where the C47 program shows its screen, and 7 (normal) when NAV ends. If you stop it with R/S or EXIT and the screen stays frozen, key **7** STO "RefLCD". dev/NAVFULL_DRAW keeps the Free42 way.

8. The ants

The ants live in NAV as on the C47, but they wait for **flag 97** (the HP-42S flag 47 is a system flag): SF 97 and press a view number, + or an arrow.

9. What is different inside (the screens are the same)

- AGRAPH draws the ALPHA register (8-pixel columns, bit 0 at the top): the fonts are strings of column bytes; the C47 drawing modes (OR, set, clear, XOR = GRMOD 0-3) are the HP-42S AGRAPH flags 34 and 35.
- PIXEL goes through a routine that converts the C47 coordinates (row 0 at the bottom) to the DM42 ones (row 1 at the top).
- Keys (GETKEY / GETKEYA) are translated to the C47 key codes; pauses use TIME; strings use XSTR, APPEND, HEAD; the date is computed (Free42 has no C47 date functions). NAV sets SIZE 100 (R00-R99).
- Built with tools/build_free42.py from the C47 programs; tested with tools/f42 (the Free42 core with the DM42 graphics, binary arithmetic). NAVFULL tested on a DM42n with the stock firmware by the author.

10. NAVLITTLE: Sun and stars

The ALMANAC screen of NAVFULL (the header line with a line under it, the same fonts) with the Sun and the 58 navigation stars (no Moon or planet rows), no box and no ants: for a calculator with little free memory. The Moon line under the table gives the % lit, the phase glyph and the age from the mean lunation (within about 14 hours of the true age). The Sun and star values are the same as in NAVFULL. If a TBL has been run (section 11), the Sun and GHA Aries come from the tables (T on the screen).

26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0										S
BODY		GHA		DEC		HC		ZN		
ARIES		229 40.8								
☉	SUN	46 25.7		S 1 24.5		11 06.4		273.7		
★37	ARCTURUS	15 27.7		N 19 02.7		24 59.7		280.2		
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★42	ANTARES	341 55.1		S 26 29.5		27 00.7		217.3		
★56	FOMALHAUT	244 53.4		S 29 28.7		10 35.9		130.0		
★53	DENEK	279 05.4		N 45 22.8		61 13.8		039.3		
★45	SHAULA	325 49.4		S 37 07.5		24 29.0		198.3		
NAUT TWI 01:20		15:01		MOON 100% ● FULL						
RISE/SET 02:09		14:12		AGE 14.9 DAYS						
MER PASS 08:11		SD 16.0								

- Load NAVINIT_LITTLE.raw, **XEQ "INIT"** (MATRICES READY: SUN STARS 2000-2050), delete INIT; load NAVLITTLE.raw, **XEQ "NAV"**. If INIT is still loaded and has not run, the first NAV runs it (flag 81).
- NAV asks DATE, UTC, LAT, LON as above and goes straight to the ALMANAC screen: ▲ / ▼ one hour later / earlier (the screen stays while it computes), + ends (GrMod 0, screen and stack cleared).
- Programs: NAVLITTLE about 10 KB, NAVINIT_LITTLE about 8 KB (.raw files), matrices 667 numbers.
- Tested in the Free42 core against the same screen in the C47 simulator, pixel by pixel (tests/test_f42_little.py); not yet on a real calculator.

11. Almanac tables: TBL_1, TBL_5 and TBL_50

- Optional: Chebyshev coefficients fitted to JPL DE421. **TBL_1**: 1 Oct 2026 – 30 Sep 2027 (8,441 numbers, file 97 KB). **TBL_5**: 1 Oct 2026 – 30 Sep 2031 (41,882 numbers, file 481 KB).
- Load ONE, **XEQ "TBL"** once: it builds the matrices TSU, TVE, TMA, TJU, TSA, TMO, TAR and sets flag 10. Then delete TBL; the matrices stay (about 135 KB for 1 year, 670 KB for 5 years).
- NAVFULL and NAVLITTLE then take the Sun, the Moon, the planets and GHA Aries from the tables inside their period (NAVLITTLE: the Sun and Aries), and the screens show **T** instead of S. Outside the period, or after **CF 10**, the series are used. The values agree with the series to 0.1'.
- The tables need more memory than the C47 has: they are meant for Free42 / Plus42 on a PC or phone, or a DM42n with enough free memory.
- **TBL_50**: 1 Jan 2000 – 31 Dec 2050 (file 4.9 MB, about 6.8 MB of matrices): Free42 / Plus42 on a PC only. Not in the package: download TBL_50.raw from the release tbl50-2000-2050 (github.com/valdesv/almanac47/releases/tag/tbl50-2000-2050) or make it with [tools/almanac/README.md](#).